

Statistical Questions and Describing Variability

Telling a statistical question from a single-answer question, measuring spread, and describing what a data set looks like

Directions:

- A **statistical question** is one you answer by collecting data that *varies*. “How tall is the flagpole?” has one answer. “How tall are the trees in the park?” has many, and they differ — that variability is what makes it statistical.
- **Spread** says how far apart the values are: $\text{range} = \text{largest} - \text{smallest}$, and the $\text{interquartile range} = \text{upper quartile} - \text{lower quartile}$.
- To find quartiles, put the values in order, split them into a lower half and an upper half, and take the median of each half.
- When you are asked to *describe* a data set, give a typical value **and** a measure of spread. One without the other is only half the story.

1. Two questions about a piano club:

Question 1: How many minutes did Ana practise yesterday?

Question 2: How many minutes do the members of the club practise each day?

Which one is a statistical question? Write its number on the answer line, and say why the other one is not.

Answer: _____

2. Seven students recorded the minutes they read last night.

12	20	15	30	18	25	22
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What is the range of these times, in minutes?

Answer: _____

3. Eight students were asked how many pets they own.

0	1	2	1	3	0	2	1
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- (a) Write a statistical question that this data could answer: _____
- (b) the median number of pets: _____

4. The high temperature was recorded for six days, in degrees Celsius.

14	19	11	22	17	20
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- (a) the lowest temperature: _____
- (b) the highest temperature: _____
- (c) the range: _____

5. Ten students scored these marks out of 10 on a spelling test.

7	8	9	8	6	10	8	7	9	8
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- (a) the mean mark: _____
- (b) the median mark: _____
- (c) Describe this data set in a sentence, using a typical value and the spread. Write it in the work space.

6. Twelve plants were measured in centimetres, already in order.

2	4	5	7	8	10	11	13	15	17	19	21
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- (a) the lower quartile: _____
- (b) the upper quartile: _____
- (c) the interquartile range: _____

7. These are the ages, in years, of the five people in one family.

4	9	12	38	41
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- (a) Is “How old is each person in this family?” a statistical question? Answer on the line and say why in the work space: _____
- (b) the range of the ages: _____

8. The Owls and the Robins each played five games. Here are their points.

Owls	20	21	22	23	24
Robins	10	16	22	28	34

Both teams have a mean of 22 points.

- (a) the range for the Owls: _____
- (b) the range for the Robins: _____
- (c) Which team is more consistent? Back your answer with the two ranges, in the work space.

9. A library counted the visitors who borrowed a book each day.

Day	Mon	Tue	Wed	Thu	Fri
Borrowers	4	7	3	9	5

- (a) the busiest day: _____
- (b) the total for the week: _____
- (c) the range of the daily counts: _____

10. Nine people at a clinic recorded how many minutes they waited.

1	2	2	3	4	4	5	6	45
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- (a) the mean wait: _____
- (b) the median wait: _____
- (c) the range: _____
- (d) Explain which of the mean and the median describes a typical wait better here, and why. Write it in the work space.